HEIRSTONE CONSULTING

CONSERVATION TOURISM IN AFRICA

The bed and the hectare

What high-value, low-volume tourism earns per bed and per hectare, who is paid for the land, and what it would take to make the model work in North Africa



THE REPORT IN SIX NUMBERS

6.2 to 6.7

beds per 1,000 ha in the Olare Motorogi and Naboisho conservancies of the Maasai Mara, about five times the Okavango Delta's 1.29 (Heirstone analysis)

US$389

tourism receipts per international arrival in Rwanda in 2019, against US$1,094 in Egypt (Heirstone analysis of World Bank data)

US$41

per hectare a year paid to Olare Orok landowners in 2011, up from US$33 when the conservancy opened in 2006

US$130

conservancy fee per guest-night in 2026 in Mara North, Olare Motorogi and Naboisho, collected through camp rates

US$2,754

revenue per hectare at Volcanoes National Park, Rwanda, in 2025: US$35.8m from about 13,000 ha (Heirstone analysis)

3 million

nature-reserve visitors a year in Egypt's protected areas, more than 20 times the visitors to Rwanda's national parks in 2024 (UNDP BIOFIN; Heirstone analysis)

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ABOUT THIS REPORT

Heirstone Consulting advises tourism authorities, land agencies, developers and investors across the Middle East and Africa on destinations, feasibility and specialised real estate. This report uses public sources current in 2026: conservancy, park-authority and operator disclosures, UNESCO and IUCN documents, government statistics, company results and published research.

Where we combine published figures to make a new one, such as beds per 1,000 hectares, revenue per hectare or receipts per arrival, the calculation is shown and labelled Heirstone analysis. Rack rates are published prices, not achieved rates. Local-currency figures are converted at the World Bank annual average rate for the year stated.

SUMMARY

High-value, low-volume tourism earns less per arrival than Egypt's volume model, and far more per hectare than any other conservation income we measured

The model pays when it is measured by land. Three numbers decide whether it works: beds per hectare, the fee charged on each guest-night, and the share of that fee paid to whoever owns the land. No country we reviewed publishes all three for all its sites.

1 **Per arrival, the flagships trail Egypt.**

In 2019 Botswana earned about US$319 per international arrival (2018 data), Rwanda US$389 and Namibia US$273, against about US$1,094 in Egypt. Among the safari economies only Tanzania, at about US$1,719, clearly beat it. Safari guests are a small share of national arrivals: the 39,374 tourists who arrived through Maun, the Okavango's air gateway, in 2023 were about 3% of Botswana's 1.18 million international tourists that year.

2 **Per hectare, they lead by orders of magnitude.**

Volcanoes National Park took in about US$2,754 per hectare in 2025. Maasai Mara landowners have been paid US$25 to US$45 per hectare a year. Namibia's communal conservancies generated about US$0.54 per hectare in cash and in-kind benefits in 2023.

3 **Density sets land yield more than price does.**

Olare Motorogi caps tourism at 94 beds on 15,200 ha and Naboisho at 150 beds on 22,500 ha, about five times the Okavango's bed density. At peak rack rates a full Mara conservancy grosses about US$7 per hectare a night; Mombo, at nearly five times the price per guest, grosses about US$2.60 to US$2.90.

4 **The lease is the asset.**

A US$130 fee on each guest-night, collected through camp rates, funds three of the main Mara conservancies. At that rate Olare Motorogi's 2012 lease bill of US$426,400 is covered at about 10% bed occupancy, and a lease at the 2011 rate equals 14% of what a full year of guest-nights would raise. Most of the value accrues to whoever holds a long lease beside a scarce attraction.

5 **North Africa already has the land and the visitors.**

Egypt's protected areas cover about 150,000 km², and its nature reserves draw about 3 million visitors a year. At a tenth of Rwanda's park revenue per visitor, fees on that draw would come to up to about US$84m a year (Heirstone analysis). Morocco's Kasbah du Toubkal adds a 5% levy for its village association, and in Saudi Arabia The Red Sea's caps work out at 933 ha per hotel key, the closest regional match to safari density.

WHAT IT MEANS

Tourism authorities should collect conservation money on the guest-night and publish the bed cap per hectare. Botswana withdrew a US$30 visitor levy in the year it was announced, according to Your Botswana.

Landowners should tie lease increases to the fee pool and hold out for terms of 15 years or more.

Investors and lenders should underwrite the lease and the cap, and treat the room rate as the variable.

North African governments could pilot one capped concession inside an existing protected area, with a per-night fee and a community levy, and publish the results.

16

sites and zones in the Heirstone Bed and Hectare Benchmark: 14 in seven African countries and two Saudi comparators

6

tests in the Heirstone Low-Density Test for land offered for high-value tourism

CHAPTER 1

01

Less per arrival than Egypt

Botswana, Rwanda and Namibia are the usual examples of high prices and few guests. In 2019 each earned under US$400 per international arrival, no more than 36% of Egypt's figure.

US$1,094

tourism receipts per international arrival in Egypt in 2019, against US$389 in Rwanda and US$273 in Namibia (Heirstone analysis of World Bank data)

Every account of high-value, low-volume tourism starts with price. Mombo, in Botswana's Okavango Delta, lists rates of US$3,062 to US$5,345 per person per night for 2026-27. Bisate, beside Rwanda's Volcanoes National Park, charges US$2,723 to US$4,023 before the US$1,500 gorilla permit. One night at either costs more than the average receipts per international arrival in any of the ten countries in Exhibit 1 in 2019.

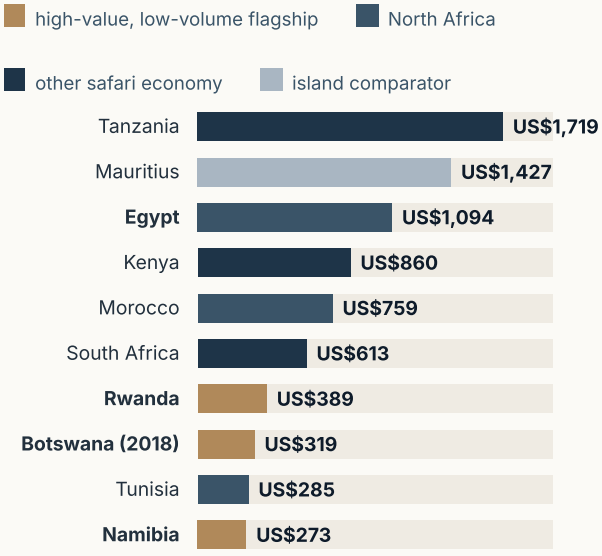
The national accounts tell a different story. In 2019 Botswana earned about US$319 per international arrival (2018 data), Rwanda about US$389 and Namibia about US$273. Egypt, selling Red Sea beach holidays and Nile heritage in volume, earned about US$1,094, and Morocco about US$759. Tanzania, at about US$1,719, was the only safari economy clearly ahead of Egypt.

Much of the gap lies in the denominator. Arrivals statistics count everyone who crosses a border: business travellers, conference delegates, family visitors and traders from neighbouring countries, as well as safari guests. In Kenya 40.8% of 2024 arrivals came from elsewhere in Africa, according to the Ministry of Tourism and Wildlife. In 2023 Maun, the Okavango's gateway, received 39,374 tourists, about 3% of Botswana's 1.18 million. Definitions differ by country, but the gap holds on travel spending alone, excluding air fares: about US$1,000 per arrival in Egypt in 2019 and US$280 in Rwanda.

EXHIBIT 1

Three high-value, low-volume flagships earned less per arrival in 2019 than Egypt

International tourism receipts per international arrival, US$, 2019 (Botswana 2018)



Source: World Bank, World Development Indicators, ST.INT.RCPT.CD and ST.INT.ARVL (from UN Tourism), accessed 2026. **Heirstone analysis.** Receipts divided by arrivals; Botswana uses 2018 receipts of US$584.2m and 1.83 million arrivals, the latest year with both series. Definitions vary by country; on travel receipts alone (ST.INT.TVLR.CD, excluding passenger transport) the 2019 figures are about US$1,000 for Egypt, US$280 for Rwanda, US$318 for Botswana (2018) and US$211 for Namibia.

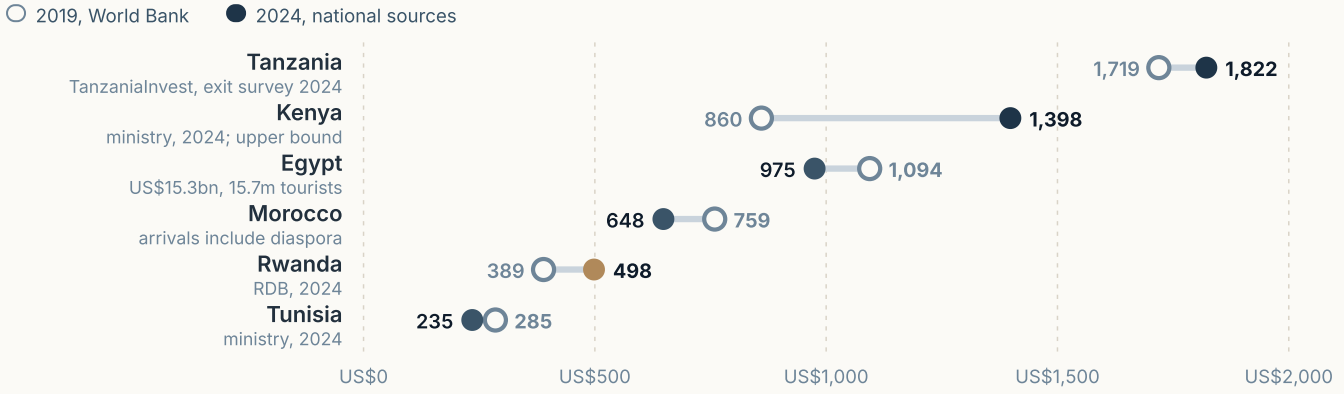
THE ENCLAVE AND THE NATIONAL AVERAGE

Five years on, Egypt still earned about twice Rwanda's receipts per arrival

EXHIBIT 2

Apart from Kenya, the order held between 2019 and 2024

Tourism receipts or earnings per international arrival, US$, 2019 (World Bank) and 2024 (national sources)



Source: World Bank WDI (2019); TanzaniaInvest reporting the International Visitors' Exit Survey 2024; Kenya Ministry of Tourism and Wildlife via Citizen Digital (Feb 2025), converted at KES 134.82 per US$ (2024); Egypt Today and WAM (2024); Finances News Hebdo (Feb 2025), MAD 9.94 per US$; Kapitalis (Feb 2025), TND 3.11 per US$; RDB via The New Times (Apr 2025). **Heirstone analysis.** 2024 national figures use different definitions and are less comparable than the 2019 series.

National figures for 2024 come from ministries and tourism boards rather than one database, so they compare less cleanly; apart from Kenya, whose ministry figure is an upper bound, the order is unchanged. Tanzania reported about US$1,822 per arrival, according to TanzaniaInvest's account of its 2024 exit survey. Egypt earned about US$975 per tourist on US$15.3bn of revenue. Rwanda rose to about US$498. Kenya's ministry total gives up to US$1,398, an upper bound because it may include some domestic spending.

Rwanda shows how a flagship sits inside a national total. Its national parks earned US$38.8m in 2024 from 138,000 visitors, about US$281 per park visitor. Total tourism revenue was US$647m from more than 1.3 million visitors, and meetings and events brought in US$84.8m, more than twice the parks. The gorilla parks carry the brand; meetings and events alone earned more than twice as much.

Judged by receipts per arrival, a conservancy or a gorilla park looks small, because its guests are a fraction of the national count. The better denominator is the land the model keeps open, and the rest of this report counts in hectares.

US$281

national-park revenue per park visitor in Rwanda in 2024: US$38.8m from 138,000 visitors (Heirstone analysis)

11%

national-park visitors against Rwanda's 2024 visitor total: 138,000 of more than 1.3 million (Heirstone analysis)

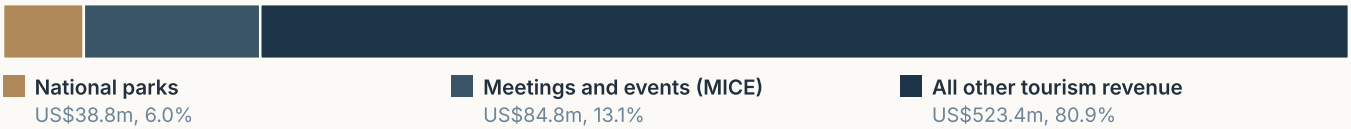
27%

of Kenya's 2024 international arrivals came for meetings, incentives, conferences and events

EXHIBIT 3

Rwanda's national parks earned 6% of its tourism revenue in 2024; meetings and events earned twice as much

Rwanda tourism revenue by source, US$ million, 2024 (total US$647m)

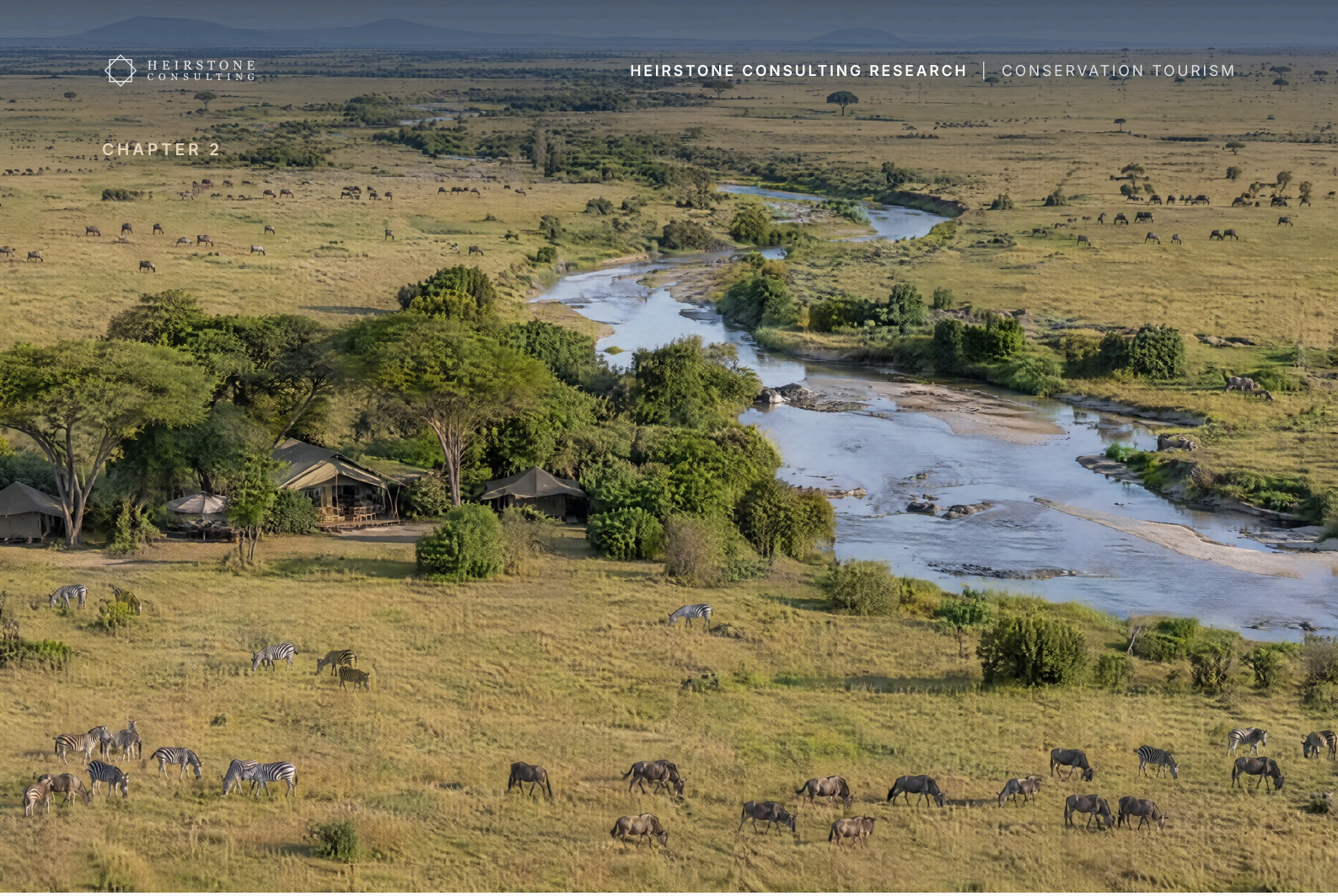


National parks
US$38.8m, 6.0%

Meetings and events (MICE)
US$84.8m, 13.1%

All other tourism revenue
US$523.4m, 80.9%

Source: Rwanda Development Board, as reported by The New Times (14 April 2025). **Heirstone analysis.** "All other" is the total less parks and MICE; it includes leisure, business and visiting-friends spending outside the parks.



02

Counting beds by the hectare

The Heirstone Bed and Hectare Benchmark uses bed caps, concession areas, fees and land payments published by conservancies, park authorities, operators, UNESCO and IUCN. Each ratio is our calculation from those inputs.

Low volume is usually described as a number of guests. For land, the measure that matters is beds per hectare, because beds drive vehicle traffic, staff housing, water use and the fee pool all at once. Few destinations publish it. The Mara conservancies are an exception. Olare Motorogi limits tourism to 94 beds in five light-footprint camps on 15,200 ha, with one game-viewing vehicle for every 2,100 acres. Mara Naboisho caps beds at 150 in eight camps on 22,500 ha.

That is 6.2 and 6.7 beds per 1,000 ha. The Okavango Delta, the usual reference point for the model, had 2,129 tourist beds in an area of 16,500 km^2 in 2014, according to an IUCN evaluation, or about 1.29 beds per 1,000 ha. Mombo's 35,000 ha private concession on Chief's Island takes up to 18 guests, according to its operator, or 0.51 beds per 1,000 ha (0.57 on its 20 berths). Both sell to the same international guests, yet their bed densities differ from the Mara's by a factor of five to thirteen.

An often-quoted rule of thumb says a premium conservancy should have no more than one bed per 700 acres (283 ha). We could not trace it to a primary source, and the Mara's own published caps are denser. Olare Motorogi works out at about 355 to 400 acres per bed, depending on which of the two areas its association publishes is used. We do not use the 700-acre figure as a benchmark; the exhibits use published caps.

THE HEIRSTONE BED AND HECTARE BENCHMARK

The benchmark sits in East and Southern Africa; the places tested in Chapter 7 lie to the north and west

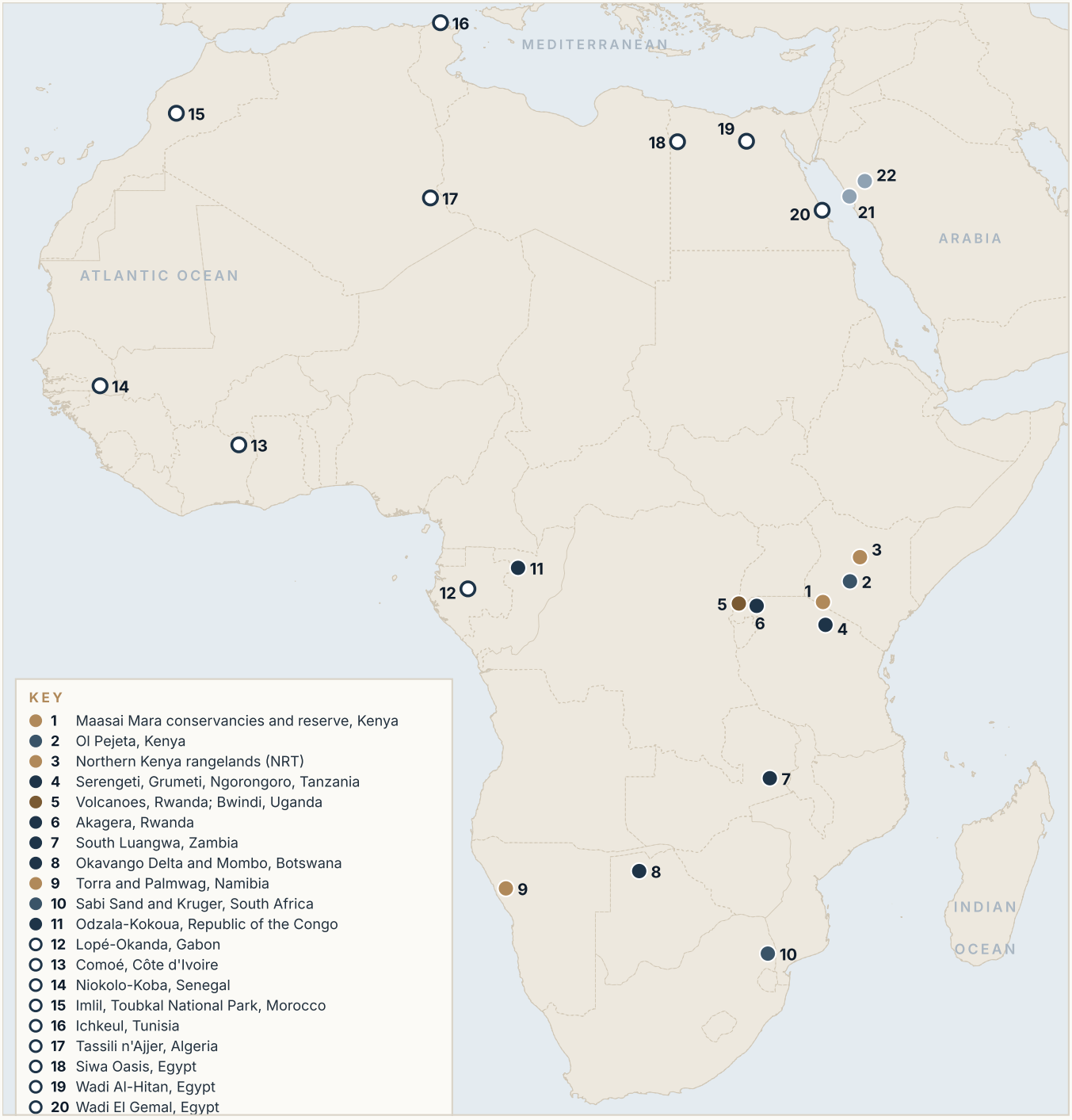
EXHIBIT 4

Every African benchmark site with a published cap lies in East or Southern Africa

Benchmark sites by land model, with the North African, francophone and Gulf sites discussed in Chapter 7; grouped sites share one marker

community conservancy · state park or concession · private reserve · permit-capped park · Gulf comparator

○ northern or francophone candidate



Source: Natural Earth (land, borders). Coordinates from UNESCO (Okavango, Comoé, Niokolo-Koba, Ichkeul, Tassili n'Ajjer, Wadi Al-Hitan) and GeoNames (Volcanoes, Odzala-Kokoua, Sabi Sand); all other positions, and grouped markers, are approximate.

THE BENCHMARK

The Heirstone Bed and Hectare Benchmark, 2026

Sites where both an area and a bed, permit or key capacity are published, or that set a reference point for prices, fees or land payments. Rack rates: published 2026 or 2026-27 prices per person per night sharing (pppn) at named camps (Kicheche in the Mara; Wilderness elsewhere). n/p: not published or not found; n/a: not applicable. Converted at World Bank annual averages: BWP 12.37 (2022), ZAR 17.89 (2025), N$ 18.45 (2023).

EXHIBIT 5

Published caps put the Mara conservancies at about five times the Okavango's density and Volcanoes at seven gorilla permits per 1,000 ha a day

Area, capacity, density, published prices, guest fees and land payments for 16 benchmark sites and zones

SITE	COUNTRY	AREA (HA)	BEDS OR CAPACITY	BEDS PER 1,000 HA	HA PER BED	RACK RATE, US$ PPPN	FEE PER GUEST	PAYMENT TO LANDOWNERS OR COMMUNITIES
COMMUNITY CONSERVANCIES (LEASED PRIVATE OR COMMUNAL LAND)								
Olare Motorogi	Kenya	15,200	94 beds, 5 camps	6.2	162	765-1,095	US$130 per night	US$41/ha (2011); US$426,400 to 217 owners (2012)
Mara Naboisho	Kenya	22,500	150 beds, 8 camps	6.7	150	765-1,095	US$130 per night	>US$900,000 (2018); c.US$45/ha
Mara North	Kenya	28,487-29,170	12 partners; beds n/p	n/p	n/p	625-995	US$130 per night	767-783 landowners (MMWCA); total n/p
Namibia, 86 conservancies	Namibia	16.6m	1,038 JV tourism jobs	n/p	n/p	632-2,267	n/p	US$0.54/ha all benefits; US$0.19/ha JV tourism wages (2023)
NRT carbon project	Kenya	1.9m	n/a	n/a	n/a	n/a	n/a	US$1.30/ha carbon distribution (2024)
STATE LAND: CONCESSIONS AND PARKS								
Okavango Delta	Botswana	1,650,000	2,129 beds (2014)	1.29	775	3,062-5,345	BWP270 (US$22) a day	24-bed cap per lodge
Mombo concession	Botswana	35,000	18 guests (20 berths)	0.51-0.57	1,750-1,944	3,062-5,345	included in rate	n/a (state land)
Mombo zone (1999 plan)	Botswana	30,000	2 lodges x 24-bed cap	1.6	625	n/a	n/a	15-year concession
Ngorongoro CA	Tanzania	829,200	n/p	n/p	n/p	n/p	US$70.80 + US$59 a night	n/a
Serengeti NP	Tanzania	n/p	n/p	n/p	n/p	998-1,617	US$60 + US$50 a night (2020/21)	n/a
PRIVATE RESERVES								
Sabi Sand	South Africa	n/p	25 lodges	n/p	n/p	n/p	R170 (US$9.50) per entry	n/a (private owners)
Ol Pejeta	Kenya	n/p	n/p	n/p	n/p	570-860 + fees	US$187 per night	n/p
PERMIT-CAPPED PARKS								
Volcanoes NP	Rwanda	c.13,000	96 gorilla permits a day	7.4 permits	n/a	2,723-4,023 + permit	US$1,500 permit	10% of revenue to communities; US$2,754/ha gross (2025)
Bwindi, Mgahinga	Uganda	n/p	n/p	n/p	n/p	n/p	US$800; habituation US$1,500	n/p
GULF COMPARATORS (KEYS, NOT BEDS)								
The Red Sea	Saudi Arabia	2,800,000	c.3,000 keys; 1m visitor cap	1.07 keys	933 per key	n/p	n/p	n/a (state-owned developer)
AlUla (2035 targets)	Saudi Arabia	2,256,100	9,400 keys; 2m visitors	4.17 keys	240 per key	n/p	n/p	n/a; 80% of county in reserves

Source: Heirstone Bed and Hectare Benchmark 2026, compiled from conservancy, park-authority, operator, UNESCO, IUCN, IFC, NACSO, NRT, RDB, Red Sea Global and RCU sources listed under Chapter 2 in the notes. Namibian per-hectare figures use the 166,179 km² managed by conservancies. Mombo: the operator states 18 guests; its suites sleep 20. **Heirstone analysis.** Ratios are our calculations.

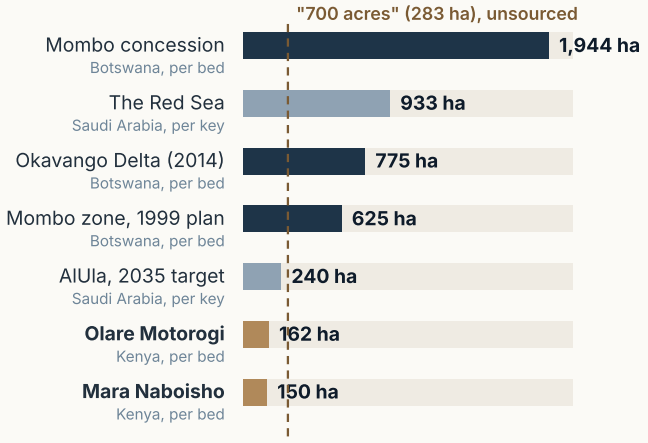
DENSITY AGAINST PRICE

A full Mara conservancy earns more per hectare each night than Mombo does at five times the price

EXHIBIT 6

The Mara conservancies give each bed a tenth or less of the land Mombo gives

Hectares of land per guest bed (or per hotel key for the Saudi comparators)



Source: Heirstone analysis. Area divided by published bed cap, guest capacity or key count. Olare Motorogi uses 15,200 ha; on the association's 33,386 acres it would be 144 ha (355 acres) per bed. Mombo uses the operator's 18 guests; on 20 berths it is 1,750 ha. Inputs: MMWCA; conservancy sites; IUCN (2014); Wilderness; IFC (1999); Red Sea Global; RCU.

Exhibit 6 turns the benchmark round: land per bed. Mombo gives each guest bed 1,750 to 1,944 ha (20 berths or 18 guests); Olare Motorogi and Naboisho give 150 to 162 ha, a tenth or less. Neither meets the 700-acre rule of thumb, but both caps are published. The Saudi destinations count keys rather than beds: once a key holds more than 1.2 beds, as most do, The Red Sea's 933 ha per key gives each bed less land than the Okavango's 775 ha.

Price runs the other way. Kicheche Bush in Olare Motorogi lists US$765 to US$1,095 per person per night for 2026, conservancy fee included. Mombo lists US$3,062 to US$5,043 for 2026, rising to US$5,345 at the 2027 peak. At the 2026 peaks Mombo charges 4.6 times as much per guest.

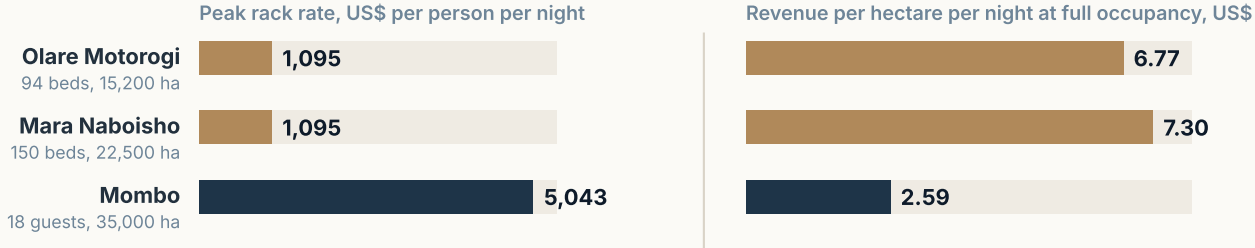
Multiply beds by price and divide by land, and density wins. A full Olare Motorogi at Kicheche's peak rate would gross about US$6.80 per hectare per night, and Naboisho about US$7.30. A full Mombo at its 2026 peak would gross about US$2.60 with 18 guests, or US$2.90 with 20. The Mara conservancies earn 2.3 to 2.8 times as much from each hectare on a peak night, and 2.7 to 3.2 times as much at the low-season rates of both camps.

The calculation is illustrative: it assumes every bed is full and priced like Kicheche, while some Mara camps charge more and all sell through agents below rack rate. The ranking survives those adjustments, because the gap in density is wider than the gap in price.

EXHIBIT 7

Density outweighs a near-fivefold price premium

Peak 2026 rack rate and implied gross revenue per hectare per night at 100% occupancy, US$



Source: Kicheche Camps 2026 rack rates (Kicheche Bush and Valley, used as proxy for each conservancy); Wilderness Mombo page (2026); Olare Motorogi and Naboisho bed caps; MMWCA areas. **Heirstone analysis.** Beds x peak rate / hectares. Illustrative: assumes one guest per bed, full occupancy and rack rates.

CHAPTER 3

03 The lease is the asset

In the Mara, camps rent the land from Maasai landowners through the conservancy, and a fee on every guest-night pays the rent.

16,530

landowners in the 24 member conservancies of the Maasai Mara Wildlife Conservancies Association, with 50 tourism partners

10.3m ha

under Kenya's conservancy movement, which says it secures 65% of the country's wildlife found outside parks and reserves

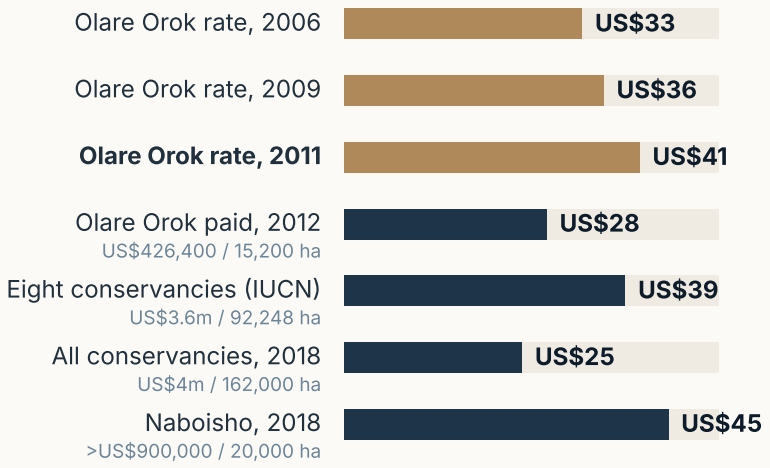
Olare Orok Conservancy, which with Motorogi now forms Olare Motorogi, opened in May 2006. Landowners leased their land to the conservancy for KES 2,500 per hectare a year, about US$33. The rate rose to US$36 in 2009 and US$41 in 2011, according to an FAO case study. After an initial lease of two and a half years, about 90% of landowners signed a 15-year contract covering 2010 to 2025.

In 2012 the conservancy paid US$426,400 to 217 landowners, about US$2,000 per family. Its management and administration costs that year were US$140,165, equal to about a third of the lease bill. Across the Mara, the conservancy association's chief executive put total lease income at about US$4m a year in 2018, on about 162,000 ha.

EXHIBIT 8

Mara landowners have been paid between US$25 and US$45 per hectare a year

Published lease rates and payments per hectare of conservancy land, US$ per year



Source: FAO, PES case study: Olare Orok Conservancy (c.2013); IUCN policy brief on payments for wildlife conservation in the Maasai Mara (undated); Mongabay commentary by MMWCA and Maliasili (Mar 2018). **Heirstone analysis.** Totals divided by stated areas. Naboisho's 20,000 ha is the 2018 figure; MMWCA now lists 22,500 ha.

The Olare Orok contract ran to 2025. We found no published terms for its successor. It is the largest gap in the public record for this model, since the length and price of the lease set what a camp can be financed against.

Neighbouring conservancies use the same instrument on different terms. Naboisho, which its own site says was formed by more than 700 landowners, paid more than US$900,000 a year in leases in 2018, according to the association's chief executive. Nashulai, founded in 2016 on about 2,400 ha, signs renewable 10-year contracts, 64 of them by late 2025 according to Mongabay. Mara North, formed in January 2009, describes its leases as the first time Maasai landowners received substantial and direct income from wildlife.

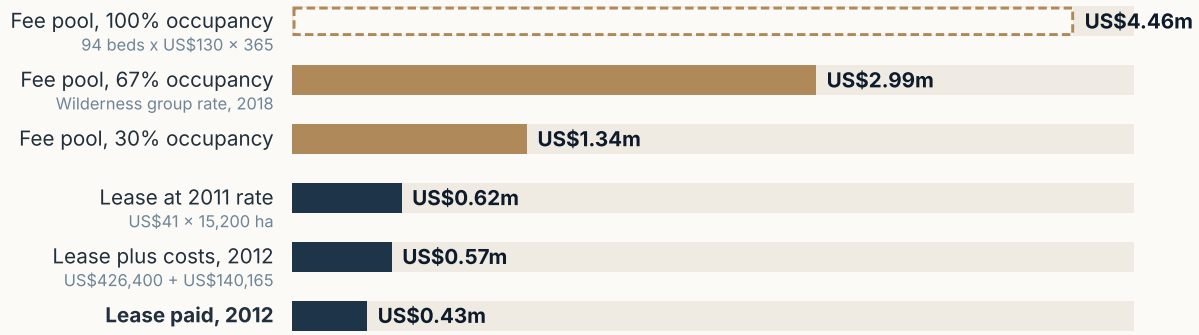
THE FEE POOL

At US$130 a guest-night, Olare Motorogi's 2012 lease bill is covered at about 10% occupancy

EXHIBIT 9

A full year of conservancy fees would be worth about seven times the lease at the 2011 rate

Olare Motorogi: conservancy-fee pool at 2026 fee levels and selected occupancies, against lease payments, US$ million a year



Source: Olare Motorogi Conservancy (94-bed cap); Kicheche Camps 2026 rack rates (US$130 conservancy fee); FAO PES case study (lease rates, 2012 payments and costs); African Financials (Wilderness 67% occupancy, H1 FY2019). **Heirstone analysis.** Fee pool = beds x fee x 365 x occupancy; assumes one adult per bed. Years differ: 2026 fee against 2011-12 lease data. Occupancy cases are illustrative; the conservancy does not publish occupancy.

The Mara conservancies fund their leases from a fee on every guest-night, bundled into camp rates. Kicheche's 2026 rate card sets it at US$130 per adult per night (US$65 per child) in Mara North, Olare Motorogi and Naboisho. Partner camps in Olare Motorogi also give US$5 per guest-night to The Maa Trust for community programmes.

Set the fee against the 94-bed cap and the lease looks small. A full year of guest-nights would raise about US$4.46m, or US$293 per hectare. The 2012 lease bill of US$426,400, about US$4,540 per bed, is covered by about 35 guest-nights per bed, roughly 10% occupancy. Add the 2012 management costs and the break-even rises to about 13%. A lease at the 2011 rate of US$41 per hectare would need 14%.

For camp operators, the lease is a small cost relative to what the land earns. For landowners, it is a fixed payment out of a pool that varies with fees and occupancy. As fees and occupancy rise, the gap between what a hectare earns and what it is paid widens, and that gap is likely to frame lease renewals.

The numbers carry caveats. The conservancy fee also pays for rangers, management and roads, not only leases. We do not know the fee level in 2012, and occupancy is not published. At the 67% occupancy Wilderness reported for its group in 2018, the fee pool would be about five times a lease at the 2011 rate; at 30%, about twice.

US$4,540

lease paid per bed in 2012 (US$426,400 across 94 beds; Heirstone analysis)

33%

management and administration costs as a share of lease payments, Olare Orok, 2012

14%

share of the full-occupancy fee pool needed to pay a lease at US$41 per hectare (Heirstone analysis)

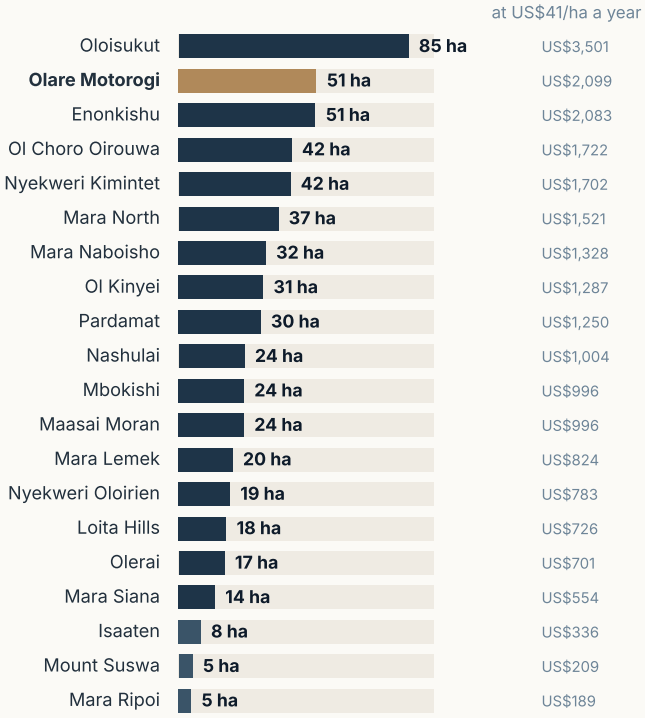
WHO RECEIVES THE LEASE

The same rate per hectare means very different incomes, because average holdings range from about 5 to 85 hectares

EXHIBIT 10

Hectares per landowner vary almost twentyfold across the Mara conservancies

Conservancy area divided by registered landowners, 20 MMWCA member conservancies, with the illustrative annual lease at US$41/ha



Source: MMWCA, Mara Conservancies page (accessed 2026). **Heirstone analysis.** Area / landowners. Uses the area and landowner figures in each conservancy's header; several descriptions on the same page give other counts (Mara Siana pays leases to over 2,000 families; Olerai was formed by 23 landowners in 2017), so ratios are indicative. Excludes Enarau (one landowner) and Olderkesi (figures unclear). Lease column applies Olare Orok's 2011 rate to every conservancy for illustration only; actual rates differ.

A flat rate per hectare pays according to how much land each family holds. Across 20 Mara conservancies, the area per registered landowner runs from about 5 ha at Mara Ripoi to about 85 ha at Oloisukut. At Olare Orok's 2011 rate, a member of Mara Ripoi would receive about US$190 a year and a member of Olare Motorogi about US$2,100.

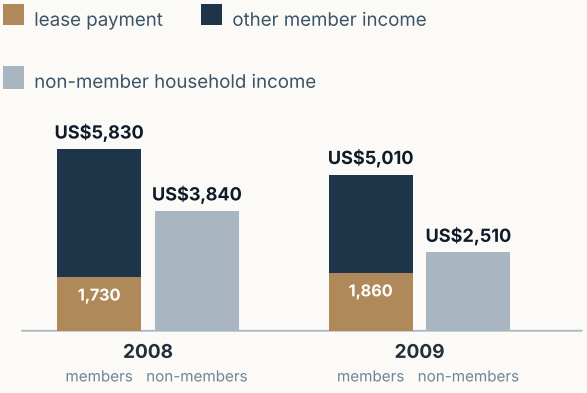
Household surveys show the lease as an addition to pastoral income. In Olare Orok, member households earned on average US$5,830 in 2008 and US$5,010 in 2009; leases supplied 30% and 37% of that, and non-member households earned US$3,840 and US$2,510. In nearby Koyiaki, a 2009-11 survey published in Pastoralism found conservancy payments averaged US$1,135 per member household, 21% of income, while livestock supplied 56% of all household income.

The same study recorded that 52% of 258 surveyed households received conservancy payments, and that a woman was the registered member in 3% of member households. Payments follow registered title, so the register decides who in a household receives them.

EXHIBIT 11

Member households in Olare Orok earned more than non-members, and leases made up 30% to 37% of their income

Mean household income, US$ per year, Olare Orok Conservancy, 2008 and 2009



Source: UCL/BEST workshop presentation, Mombasa (June 2013), citing a household survey; Pastoralism 7:10 (2017).

CHAPTER 4

04

Four orders of magnitude

Measured per hectare, Africa's conservation land produces anything from a few cents to a few thousand dollars a year. The range is wide enough that averages mislead, and it explains why the same label covers very different economies.

At the top sits Volcanoes National Park in Rwanda. It earned US$35.8m in 2025 on about 13,000 ha, about US$2,754 per hectare, from a product rationed to 96 gorilla permits a day at US$1,500 each. In the middle are the Mara conservancies, where landowners are paid US$25 to US$45 per hectare a year. At the lower end are the very large areas, where income is spread across millions of hectares: Namibia's communal conservancies, northern Kenya's carbon project and Egypt's protected areas, each well under US$2 per hectare.

Low yields per hectare can still carry large outcomes. Namibia's conservancies put 20% of the country under community management and helped elephant numbers rise from about 7,600 in 1995 to about 23,600 in 2016, according to the national association. Beds and permits concentrate value on small areas. Income spread across millions of hectares comes to cents per hectare unless another source, such as carbon, adds to it.



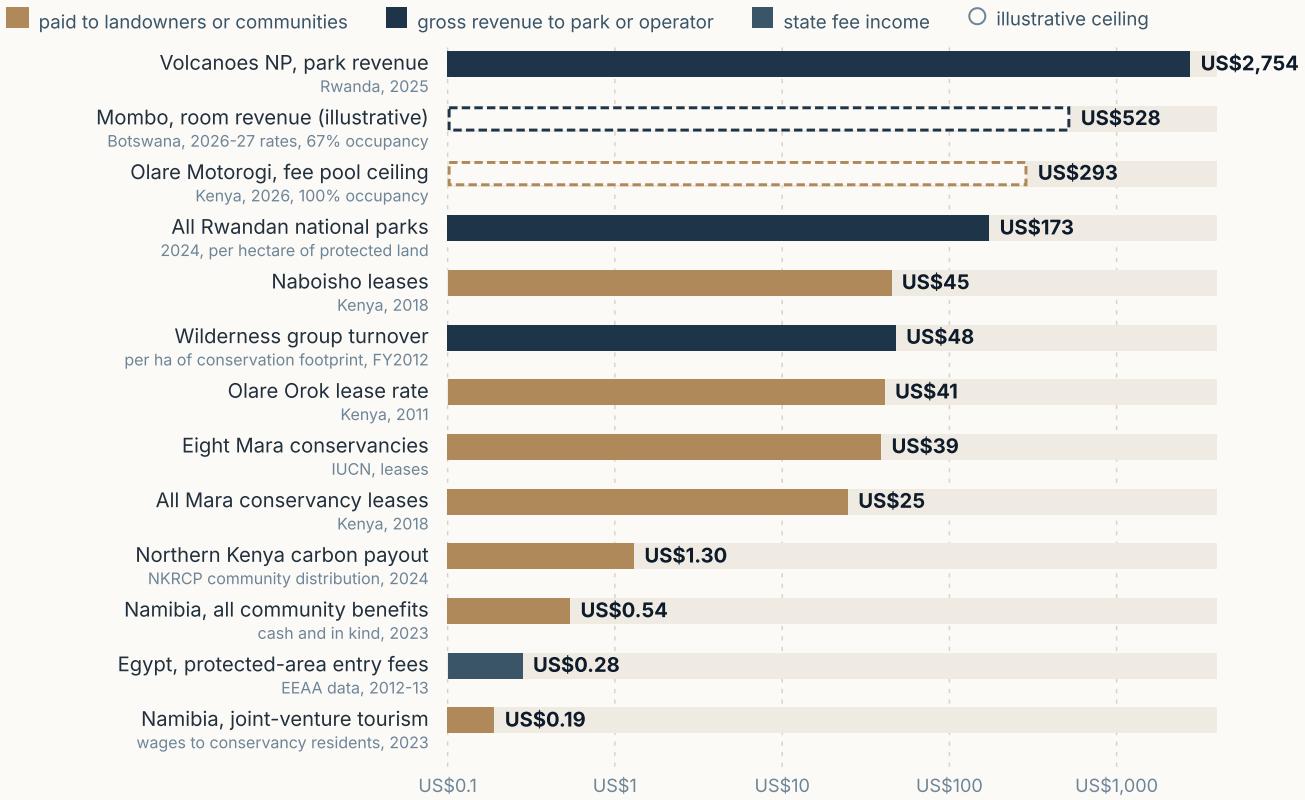
THE PER-HECTARE LADDER

From under twenty cents to more than two thousand dollars: what a hectare of conservation land earns in a year

EXHIBIT 12

Money per hectare spans four orders of magnitude across Africa's conservation land

US$ per hectare per year, logarithmic scale; colours distinguish who receives the money



Source: Heirstone analysis. Inputs and sources: RDB via Citizen Digital (2026) and Afrik21 (Volcanoes c.13,000 ha, 2022); Wilderness Mombo page with 18 guests x 365 × 67% x US$4,200 mid-rate / 35,000 ha; Olare Motorogi 94 beds x US$130 × 365 / 15,200 ha; RDB via New Times (US$38.8m, 2024) / WDI protected area 224,497 ha; Mongabay 2018; Wilderness Integrated Report 2012 (BWP1,066m / 3.2m ha at BWP 6.97); FAO; IUCN; NRT Annual Report 2024 (US$2.47m / 1.9m ha); NACSO SOCC 2023 (N$166.9m and N$57.7m / 166,179 km² at N$18.45); EEAA (2013) (LE25m / 15m ha at LE6.06). Measures differ: compare within a colour.

The ladder mixes measures on purpose, and the colours keep them apart. Gold bars are money that reaches landowners or communities. Navy bars are gross revenue to a park authority or an operator, before costs. Two bars are outlines because they are ceilings built from rack rates, not observed figures.

Three readings matter for investors and governments. First, within the gold bars, the Mara leases sit about 30 times above the northern Kenya carbon payout and about 70 times above all community benefits per hectare in Namibia's conservancies. The difference reflects bed density next to a wildlife spectacle. Second, the navy bars show how much of the value stays with the operator or the park: a Mombo-type concession grosses an illustrative US$528 per hectare a year, more than ten times what Mara landowners receive. Third, Egypt's protected areas already receive about 3 million nature-reserve visitors a year, according to UNDP. On EEAA data, entry fees collected at seven areas came to about US$0.28 per hectare of the whole estate in 2012-13, so that draw was not yet priced by the hectare.

BREADTH AND INTENSITY

Namibia shows the model spread across a fifth of a country; Rwanda shows it at its most intense

Namibia: communal conservancies

Namibia had 86 registered communal conservancies in 2023, managing 166,179 km², or 20.2% of the country, and about 244,587 people live in them, according to the national association. They generated N$166.9m (about US$9.0m at N$18.45 per US$) in cash income and in-kind benefits in 2023, up from N$140.3m in 2022. Tourism supplied N$111.2m and conservation hunting N$49.0m.

Residents earned N$90.5m (US$4.9m) in wages from conservancy enterprises, N$57.7m of it from joint-venture tourism, in which a lodge operator partners with the conservancy; the 74 joint-venture agreements employed 1,038 people. Conservancies distributed N$28.0m in cash benefits. The association estimates that community conservation added N$15.468bn (nominal) to net national income between 1990 and 2023, with an economic internal rate of return of 18%.

Per hectare these sums are small: about US$0.54 in all benefits and US$0.19 in joint-venture tourism wages. Yet the lodges inside them sell at safari prices. Wilderness Damaraland Camp, a joint venture with Torra Conservancy since the 1990s, lists US$632 to US$1,109 per person per night for 2026-27; Hoanib Skeleton Coast Camp, in the Palmwag Concession run jointly with three conservancies, lists US$1,278 to US$2,267. Namibia's lodges command safari prices; its income per hectare is low because it is spread across 16.6 million ha.

20.2%

of Namibia managed by communal conservancies, 2023

2,340

jobs facilitated by community conservation, 2023

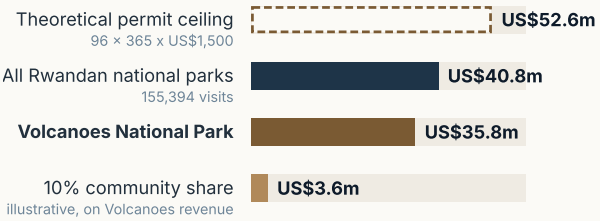
Rwanda: one product, rationed

Rwanda issues eight permits a day for each of 12 habituated gorilla families in Volcanoes National Park: 96 permits at US$1,500. In 2025 the park earned US$35.8m, 88% of the US$40.8m earned by all national parks from 155,394 visits. That equals 68% of what 96 permits a day would gross at the full foreign rate; the park sells other activities, and resident and regional visitors pay less, so the ratio is a rough guide only.

EXHIBIT 13

Volcanoes earned the equivalent of 68% of its gorilla-permit ceiling in 2025

Volcanoes National Park revenue against theoretical permit income, US$ million, 2025



Source: Visit Rwanda (RDB), gorilla tracking; RDB figures via Citizen Digital (20 Aug 2026). **Heirstone analysis.** Ceiling = 96 permits x 365 days x US$1,500. Community share applies the 10% rule to Volcanoes revenue for illustration.

Ten per cent of tourism revenue goes to communities around the parks; the development board reports more than RWF 23bn (about US$16m at the 2025 average rate) invested in nearly 1,300 projects. Uganda prices its gorilla permit at US$800, 53% of Rwanda's, while charging US$1,500 for a half-day habituation experience.

7.4

gorilla permits per 1,000 ha a day in Volcanoes, a little above the Mara conservancies' bed density (Heirstone analysis)

US$2,754

Volcanoes revenue per hectare in 2025 (Heirstone analysis)

CARBON ON THE RANGELANDS

Carbon reaches land that beds cannot, at about a thirtieth of the Mara's yield per hectare

EXHIBIT 14

Carbon paid northern Kenya's conservancies about US$1.30 per hectare in 2024, and issuance or sales have been halted twice since 2023

Northern Kenya Rangelands Carbon Project: selected events, 2012 to 2025



Source: Northern Kenya Rangelands Carbon Project website (2026); NRT Annual Report 2024; Mongabay (Jan 2025); Drilled (2026); Survival International (May 2025). Status after the 2025 review not verified.

Northern Kenya's community conservancies cover far more land than the Mara's. The Northern Rangelands Trust reports 47 member community conservancies (45 in its 2024 annual report) on 6.37 million ha in northern and coastal Kenya and in Uganda. Tourism there has generated KES 1.12bn (US$8.68m) over the trust's lifetime, according to its 2024 annual report.

Carbon has brought in more. The Northern Kenya Rangelands Carbon Project covers 1.9 million ha, started in December 2012 and reports more than 3.2 million verified credits. NRT states that KES 3.6bn (US$28.2m) had been secured by 2024, more than three times its lifetime tourism figure, and that KES 319.1m (US$2.47m) was distributed to participating conservancies in 2024. That is about US$14.8 per hectare in total since the start, and US$1.30 per hectare in 2024, against about US$39 a year for Mara leases.

The project has also been under legal and registry scrutiny. In 2023 Verra suspended issuance after a report by Survival International, and later lifted the suspension. On 31 January 2025 the Environment and Land Court at Isiolo ruled that the Biliqo Bulesa and Cherab conservancies had been established without adequate public participation and ordered them to stop operating, according to Mongabay; NRT's chief executive said the Biliqo Bulesa community had supported the carbon project. About a fifth of the project's credits come from Biliqo Bulesa. Verra then placed the project under review; according to Drilled, credits from its grazing programme may not be sold until the review is complete. We found no published credit sale prices.

For investors, the main issue is the land record. Buyers of credits, like operators signing leases, need evidence of consent and public participation behind each conservancy. For governments and community bodies, carbon is a top-up for land that beds cannot reach, worth around a thirtieth of the Mara lease rate per hectare in 2024.

US$1.30

carbon money distributed per project hectare in 2024: US$2.47m across 1.9m ha (Heirstone analysis)

US$14.8

carbon money secured per project hectare in total, 2012 to 2024: US$28.2m (Heirstone analysis)

US$39

Mara conservancy lease income per hectare per year, from the IUCN figures (Heirstone analysis)

1/5

of the project's credits originate in Biliqo Bulesa conservancy, according to Mongabay

CHAPTER 5

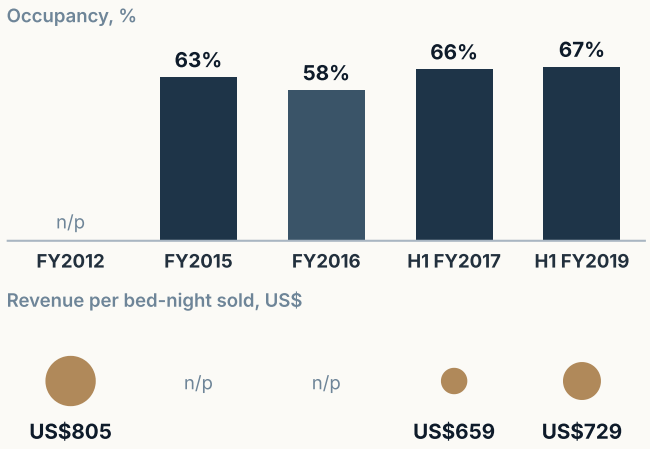
05 What the operator earns, and what the guest pays

The best-documented safari operator earned between about US$660 and US$800 per bed-night sold from 2012 to 2018. Fees are the part of the bill that governments control.

EXHIBIT 15

Wilderness earned US$660 to US$800 per bed-night while occupancy moved between 58% and 67%

Wilderness Holdings: group occupancy and group revenue per bed-night sold, selected periods



Source: Wilderness Holdings Integrated Report 2012; Tourism Update (May 2016); Mmegi (Oct 2016); African Financials (Oct 2018). **Heirstone analysis.** Group revenue / bed-nights sold, converted at BWP 6.97 (2011 and 2012 averages weighted to the March-February year), 10.90 (2016) and 10.20 (2018) per US$. Group revenue includes travel and aviation, so camp-only yield is lower. n/p: not published.

30%
trading margin, six months to August 2018, up from 26%

5·5
group employees per occupied bed, 2012 (Heirstone analysis)

Wilderness, the operator of Mombo, Bisate, Damaraland and Hoanib, published its results for several years up to 2018. In the six months to August 2018 it sold 104,855 bed-nights of 156,788 available, an occupancy of 67%, on revenue of P780m and a trading profit of P235m, a margin of 30%. The available bed-nights imply about 860 beds across the group.

Revenue per bed-night sold was about US$800 in the year to February 2012, US$660 in the six months to August 2016 and US$730 in the six months to August 2018. In pula it rose from about P5,600 to P7,400. Occupancy fell from 63% to 58% in the year to February 2016 and recovered to 66% in the six months to August 2016 as Ebola fears subsided.

The bed also carries jobs. In its 2012 year the group employed 2,847 people for 190,000 bed-nights sold, about 5.5 employees per occupied bed across camps, head office, travel and aviation. A University of Cape Town study of 16 high-end lodges in six southern African countries found 683 staff earning on average US$278 a month and supporting 4,781 dependants.

Community payments were a small share of turnover: BWP 8.5m (about US$1.2m) in its 2012 year, about 0.8% of BWP 1,066m (Heirstone analysis), with up to BWP 4.7m spent on conservation. In the Mara, landowners are paid through a separate fee on each guest-night.

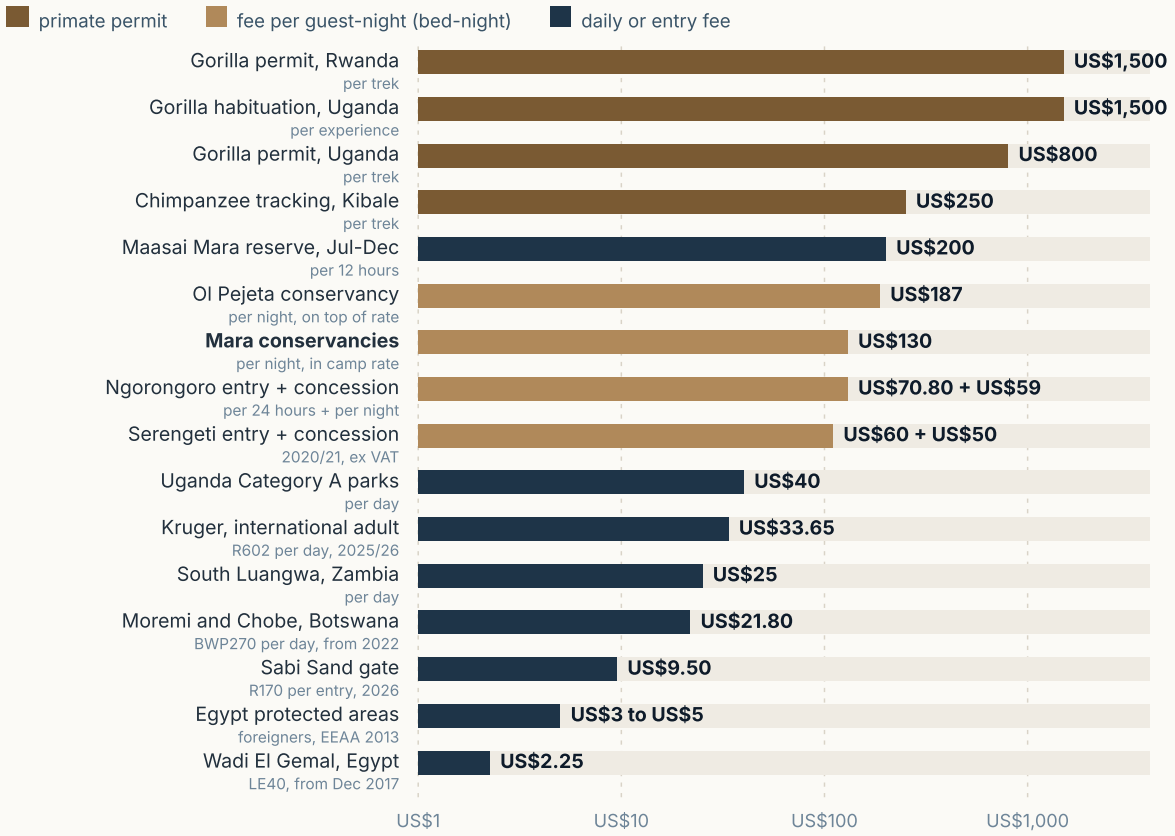
FEES

Three Mara conservancies collect US$130 a guest-night via the camp, money that pays for the land

EXHIBIT 16

Fees per foreign adult range from about US$2 to US$1,500, and the fees charged per night sit with the land

Published non-resident adult fees, US$, logarithmic scale; 2026 unless stated



Source: Visit Rwanda; UWA tariff 2024-26; Nation (2026); Kicheche 2026 rates; NCAA tariffs; TANAPA 2020/21; ATTA (Oct 2025); Zambia Tourism; Tourism Update (Feb 2022); Sabi Sand Nature Reserve; EEAA (2013); Egypt Independent (Dec 2017). Converted at ZAR 17.89 (2025), BWP 12.37 (2022), LE17.78 (2017) per US$. Bars for combined fees show the sum.

Fees are the price lever governments control directly. Botswana raised non-resident entry to Moremi and Chobe from BWP120 to BWP270 a day (about US$10 to US$22 at the 2022 rate) from April 2022, the first review in more than 20 years. UNDP's BIOFIN reports that the new fees earned US$7.8m in their first year, against US$1.8m projected.

The Mara reserve shows the limit. Business Daily reported in May 2026 that entry fees rose up to threefold in 2024, that visitor numbers fell for a second year, and that Amboseli overtook the Mara in 2025. The reserve now charges US$200 per 12-hour ticket in high season. One reading is that price rises hold best where access is already rationed.

Botswana's flat arrival levy did not proceed: a US$30 levy on non-SADC visitors was announced in May 2017 and the bill withdrawn in December 2017 for further stakeholder engagement, according to Your Botswana. Fees charged per night through the camp, as in the Mara conservancies, Ol Pejeta and Ngorongoro, are paid by the guest who uses the land and flow to whoever controls it.

In December 2017 Egypt added Wadi El Gemal, near Marsa Alam, to its fee-charging protected areas, at LE40 for foreigners (about US$2.25 at the 2017 rate) and LE5 for Egyptians; the director of Red Sea Protected Areas expected LE50m in the first year. The 2013 strategy had recorded foreign entry fees of US$3 to US$5 at other areas. A fee set in local currency can hold its dollar value if it is reviewed on a fixed timetable.

CHAPTER 6

06

Four ways to hold the land

The benchmark sorts into four land models and one Gulf variant. Each answers the same three questions differently: who holds the land, who sets the cap, and who is paid for it.

EXHIBIT 17

The four African models and the Gulf variant differ on tenure, on who sets the cap and on who is paid

Heirstone typology of high-value, low-volume land models, with benchmark evidence

	COMMUNITY CONSERVANCY	STATE CONCESSION	PRIVATE RESERVE	PERMIT-CAPPED PARK	GULF VARIANT: DEVELOPER DESTINATION
Benchmark examples	Olare Motorogi, Naboisho (Kenya); Namibian communal conservancies	Okavango and Mombo (Botswana); Serengeti, Grumeti (Tanzania)	Sabi Sand (South Africa); Ol Pejeta (Kenya)	Volcanoes (Rwanda); Bwindi (Uganda)	The Red Sea; AlUla (Saudi Arabia)
Who holds the land	Individual or communal owners, pooled by lease	The state	Private owners or an association	The state	State-owned developer or commission
Who sets the cap	Conservancy with its partner camps (94 and 150 beds)	Management plan (24 beds per lodge in Okavango reserves)	Owners; not published	Daily permits (96 a day)	Developer: keys and visitor caps
How land is paid	Lease per hectare (US$25-45 in the Mara) or JV fees	Concession fees and park fees to the state	Owners earn through lodges and gate fees	10% of revenue to communities	None published
Tenure	15-year leases (Olare Orok 2010-25); 10-year renewable (Nashulai)	15-year concession (Mombo, 1999); 25-year management (Odzala)	Private title	State park	Held by the developer
Main risk	Renewal and the fairness of the register	Tender terms and renewal; terms not published	Few beneficiaries beyond owners	One species, one price	Density creeps as keys are added
Fit for North Africa	Where families or communities hold land under customary tenure	Where protected areas are state land (most of Egypt)	Limited	Single-attraction sites	Already in use in Saudi Arabia

Source: Heirstone analysis. Typology built from the benchmark (Exhibit 5) and sources cited in Chapters 2 to 5, plus African Parks (Odzala-Kokoua 25-year agreement, 2010) and Mongabay (Nashulai, Dec 2025).

The community conservancy pays landowners directly and publishes the most. It also carries the most negotiation: leases expire, holdings are uneven, and registers decide who is paid. **The state concession** keeps land public and sells time on it; the Okavango's 24-bed cap per lodge is written into a management plan, yet we found no concession fees published.

The private reserve concentrates decisions and returns with a small number of owners. Sabi Sand has been unfenced with Kruger since 1993 and lists 25 lodges, but publishes no bed total. **The permit-capped park** gives the highest yield per hectare we found, from one rationed product, and shares 10% of revenue in Rwanda's case.

The Gulf variant fits none of the four. The Red Sea and AlUla sit on land held by a state-owned developer or a royal commission, with published caps on keys and visitors. Tenure is therefore settled from the start. That leaves the fee and the land payment, the two lines that make the African models pay for conservation, to be designed.

For North Africa, most protected land is state land, so the state concession is the natural starting point; where families or communities hold land under customary tenure, the conservancy lease is the precedent. The test on the next page applies to all of them.

A HEIRSTONE FRAMEWORK

The Heirstone Low-Density Test: six questions for any land offered for high-value tourism

Ministries, land agencies and master developers across Africa and the Middle East now describe projects as low-density, regenerative or high-value. The evidence in this report suggests the label means little unless six conditions hold. Each test below is tied to a measured case in the benchmark, and each threshold is drawn from a working model. Olare Motorogi comes closest: it meets four outright and two in part, because its 15-year lease ended in 2025 and its published land payments stop in 2012.

A site that passes all six can be priced and financed like a Mara conservancy or an Okavango concession. A site that passes the first two but fails tenure, fee and land line may attract a lodge, but the value is likely to stay with the operator and the land will not be paid to stay open. Chapter 7 applies the test to North African and Gulf sites; every North African site meets the first, a draw, at least in part.

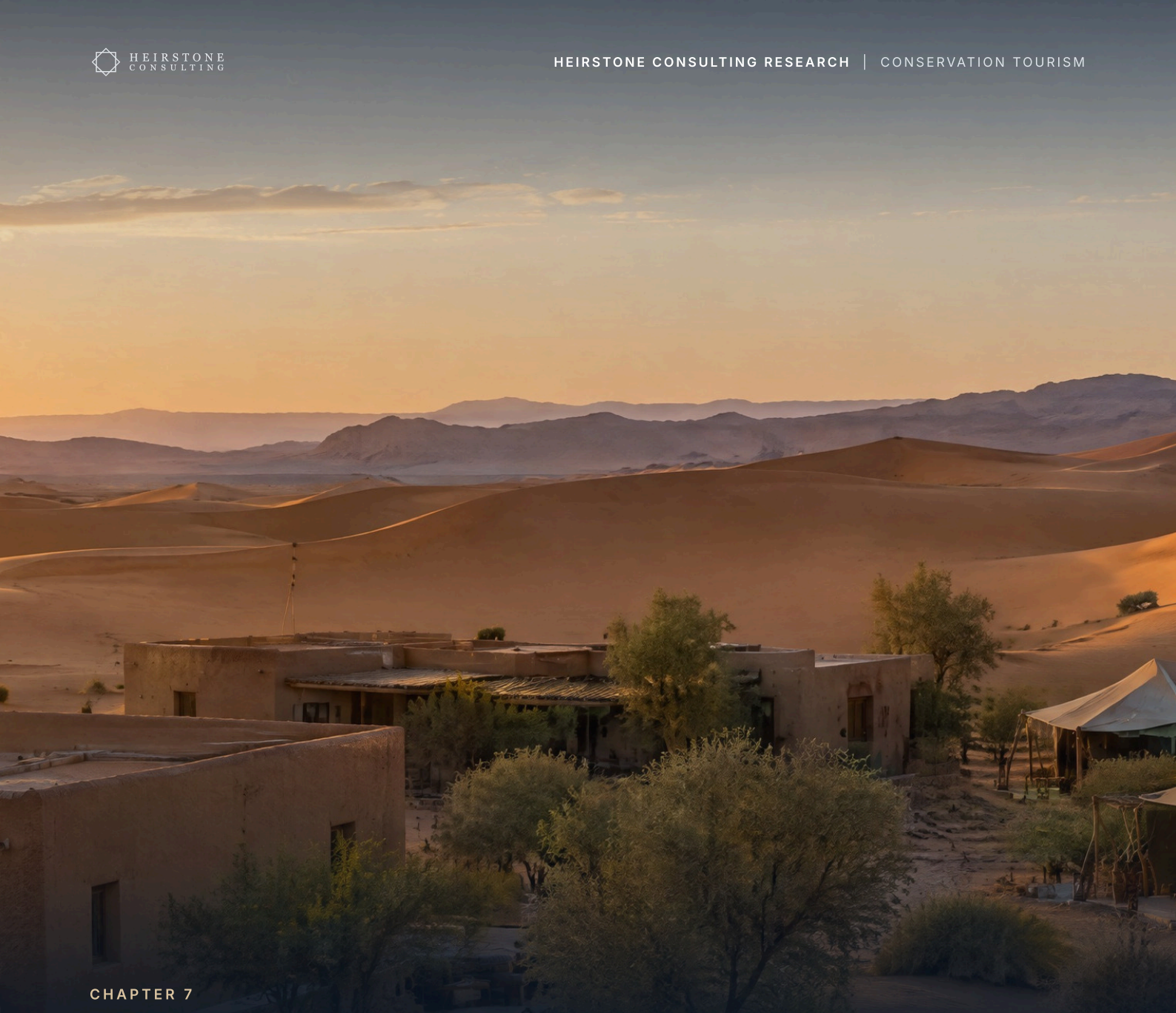
EXHIBIT 18

The Heirstone Low-Density Test sets six thresholds drawn from working models in the benchmark

Tests, evidence from the benchmark and thresholds

1	**Draw**	**Is there an attraction tied to this land that guests cannot get elsewhere on the same trip?** The top per-hectare yields sit beside the Mara migration and the Volcanoes gorillas.	Threshold **Named, place-bound draw**
2	**Cap**	**Is a bed, permit or key cap published per unit of land?** Olare Motorogi 94 beds on 15,200 ha; Naboisho 150 on 22,500 ha; Volcanoes 96 permits a day; Okavango 24 beds per lodge.	Threshold **Published cap; 7 beds or fewer per 1,000 ha**
3	**Tenure**	**Does an operator hold an exclusive right long enough to finance camps?** Olare Orok 15-year lease (2010-25); Mombo 15-year concession (1999); Odzala 25-year agreement (2010).	Threshold **15 years or more**
4	**Bed-night fee**	**Is conservation money charged per guest-night and collected through the camp?** US$130 in Mara conservancies; US$187 at Ol Pejeta; US$59 concession fee at Ngorongoro.	Threshold **Fee per guest-night, ring-fenced**
5	**Land line**	**Do landowners or communities receive a payment per hectare or a fixed share of revenue?** US$25-45/ha in the Mara; 10% of park revenue in Rwanda; 5% of invoices at Kasbah du Toubkal.	Threshold **Per-hectare payment or fixed revenue share**
6	**Disclosure**	**Are beds, fees and land payments published each year?** NACSO and RDB publish annual figures; MMWCA publishes areas and landowners.	Threshold **Annual public report**

Source: Heirstone analysis. Thresholds derived from the Heirstone Bed and Hectare Benchmark (Exhibit 5) and Chapters 3 to 5. The 7-beds-per-1,000-ha ceiling is the densest capped conservancy in the benchmark (Naboisho, 6.7), not an ecological limit.



CHAPTER 7

07 How far the model travels

North Africa has deserts, mountains, oases and marine parks that already draw millions of visitors, and in Egypt about 150,000 km^2 of protected areas. In Egypt's 2013 figures, entry fees were the main income; per-night fees, concessions and caps are the next step. Sub-Saharan Africa beyond the safari core, including the large francophone parks, has the land and, on the published record, few beds. Saudi Arabia has written caps into its new destinations.

This chapter tests what would have to change in each, using the six tests of Chapter 6.

NORTH AFRICA

Egypt's nature reserves draw about 3 million visitors a year, a base for per-night fees

Egypt

Egypt's 30 protected areas covered about 150,000 km², some 15% of its territory, according to the environment agency's 2013 financing strategy; the World Bank now records 13.2% of land as protected, close to Kenya's 14.0%. The same strategy set out the starting point for pricing: foreign entry fees of US$3 to US$5, and more than LE25m (about US$4m at the 2012 average rate) from fees at seven areas, 80% of protected-area income. Concessions, at about 3%, had clear room to grow.

Fee collection has since been extended: in December 2017 Wadi El Gemal, near Marsa Alam, announced fees of LE40 for foreigners (about US$2.25) and LE5 for Egyptians. UNDP's BIOFIN now counts 31 protected areas, about 3 million visitors a year, more than 20 times the 138,000 who visited Rwanda's national parks in 2024, and a financing gap of about US$292m for 2024-2030. If those visitors paid a tenth of Rwanda's 2024 park revenue per visitor, about US$28, fees would raise about US$84m a year, twice the gap's annual average (Heirstone analysis; an illustrative upper bound, as Egyptians pay lower fees).

The product exists. Wadi Al-Hitan, a 20,015 ha World Heritage property in Wadi El-Rayan, permits no vehicle access, according to UNESCO, and has limited on-site accommodation, which works as a cap. Adrère Amellal at Siwa runs 40 rooms without electricity. A published cap per hectare, a per-night fee and a land payment would complete the model at either site.

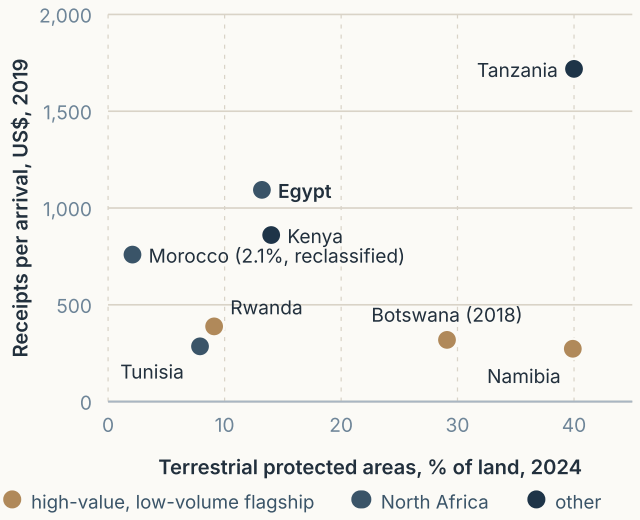
Morocco, Tunisia and Algeria

Kasbah du Toubkal at Imlil, inside Toubkal National Park and 60 km from Marrakech, adds 5% to every invoice for the Imlil village association and Education For All: the region's clearest land line. Morocco earned about US$759 per arrival in 2019 and US$648 in 2024, with arrivals including Moroccans resident abroad. Tunisia earned about US$285 per arrival in 2019 and US$235 in 2024; Ichkeul (12,600 ha) is its natural World Heritage site. Algeria's Tassili n'Ajjer covers 7.2 million ha, where, UNESCO notes, visitor groups are always accompanied by an official guide.

EXHIBIT 19

Protected land shows no clear link with receipts per arrival across the eight countries

Terrestrial protected areas as % of land (2024) against tourism receipts per arrival (2019)



Source: World Bank WDI, ER.LND.PTLD.ZS (UNEP-WCMC), ST.INT.RCPT.CD, ST.INT.ARVL. **Heirstone analysis.** Morocco's protected share fell from 30.8% (2019) after a WDPA reclassification.

150,000 km²

of protected areas in Egypt across 30 sites, some 15% of its territory (EEAA, 2013)

US$84m

a year in illustrative fee income if Egypt's 3 million reserve visitors paid a tenth of Rwanda's 2024 park revenue per visitor, twice the annual financing gap (Heirstone analysis)

5%

levy on every Kasbah du Toubkal invoice for the Imlil village association and Education For All

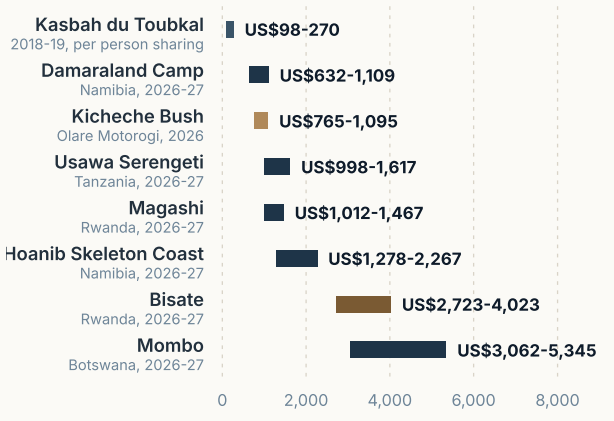
PRICE, TESTS AND THE GULF

A High Atlas eco-lodge lists rates at a quarter or less of a Mara camp's, and every North African site at least partly meets the first of the six tests, a draw

EXHIBIT 20

Kasbah du Toubkal's 2018-19 rates were a quarter or less of a Mara camp's 2026 rate and a tenth or less of the top safari camps

Published rack rates, US$ per person per night sharing



Source: Kasbah du Toubkal price list 2018-19 (EUR 175-480 per room B&B, converted at EUR 0.89 per US$, 2019, and halved per person); Kicheche 2026; Wilderness camp pages (2026). Adrère Amellal does not publish rates.

EXHIBIT 21

Every North African site has at least a partial draw; the Saudi sites score on cap and tenure

The Heirstone Low-Density Test applied to northern and Gulf sites

SITE	DRAW	CAP	TENURE	BED-NIGHT FEE	LAND LINE	DISCLOSURE
Olare Motorogi (ref.)	●	●	◑	●	●	◑
Imlil, Morocco	◑	○	○	○	●	◑
Siwa, Egypt	◑	○	○	○	○	○
Wadi El Gemal, Egypt	◑	○	○	◑	○	◑
Wadi Al-Hitan, Egypt	●	◑	○	○	○	◑
Ichkeul, Tunisia	◑	○	○	○	○	◑
Tassili n'Ajjer, Algeria	●	◑	○	○	○	○
The Red Sea, Saudi Arabia	◑	●	●	○	○	◑
AlUla, Saudi Arabia	●	◑	●	○	○	◑

● met on published evidence ◑ partly met ○ not found in published sources

Source: Heirstone analysis. Assessment from published sources cited in this report; a hollow mark means we found no published evidence, not that the condition is absent.

Price follows the tests. Kasbah du Toubkal's 2018-19 rates of EUR 175 to EUR 480 per room are about US$98 to US$270 per person sharing (a current listing shows EUR 200 to EUR 1,080 per room). The safari camps in Exhibit 20 list US$632 to US$5,345 per person, each on exclusive land beside a scarce draw.

Saudi Arabia shows another route. Red Sea Global limits development at The Red Sea to less than 1% of its 28,000 km², with about 3,000 keys in 18 hotels and a cap of 1 million visitors a year: 933 ha per key, against the Okavango's 775 ha per bed. AlUla's plan for 9,400 keys by 2035 on 22,561 km², 80% of it nature reserve, gives 240 ha per key. Both pass on cap, and on tenure because a single state-owned body controls the land for the long term. Neither publishes a per-night conservation fee or a land payment; those two lines would have to come from the African models.

North Africa's sites start from their draw. Wadi Al-Hitan and Tassili n'Ajjer, both World Heritage properties, control access, a partial cap; Wadi El Gemal has an entry fee; Imlil has a land line through the Kasbah levy. A published bed cap, a long-term concession and a per-night fee are the next three steps.

SUB-SAHARAN AFRICA, ANGLOPHONE AND FRANCOPHONE

The safari core needs lease renewals and disclosure, while the francophone parks can set terms before the first beds are built

East Africa

Kenya has the most developed market in our set for leasing land to conservation: more than 10.3 million ha in conservancies, 24 Mara conservancies with 16,530 landowners, and NRT member conservancies on 6.37 million ha in Kenya and Uganda. The Olare Orok lease ran to 2025, the reserve's 2024 fee rise coincided with falling visits, and few conservancies publish a bed cap. Tanzania earned the highest receipts per arrival in our set and already charges per night: US$50 in the Serengeti (2020/21) and US$59 at Ngorongoro. Grumeti's 350,000 acres (about 141,600 ha) became a state game reserve in 1994, with tourism run by Singita since 2006. Uganda prices gorilla permits at US$800.

Southern Africa

Botswana wrote a 24-bed cap per lodge into the Okavango plan, and its 2022 fee review earned more than four times the projected revenue; its concession rents are unpublished and the latest Delta bed count we found is from 2014. Namibia has 20.2% of the country in communal conservancies and lodge prices to match the Mara, but its income is spread across 16.6 million ha. Sabi Sand runs the private-reserve model with 25 lodges, unfenced with Kruger since 1993. Zambia charges US$25 a day at South Luangwa.

Rwanda and francophone Africa

Rwanda, a member of the Francophonie, pairs the gorilla permit with park partnerships: at Akagera, co-managed by African Parks since 2010, revenue covered operating costs in full in 2025. Comoé in Côte d'Ivoire (1,149,450 ha) left UNESCO's danger list in 2017 and Niokolo-Koba in Senegal (913,000 ha) in July 2024; Lopé-Okanda in Gabon covers 511,991 ha. Odzala-Kokoua in the Republic of the Congo (13,546 km^2) has a 25-year management agreement signed in 2010. We found no published bed caps, per-night fees or land payments for these parks, so the first concessions can set all three.

EXHIBIT 22

No country in the set publishes beds per hectare, the guest-night fee and the land payment for all its sites

What each country publishes on the three numbers, and the first priority

COUNTRY	LAND MODEL	PUBLISHED NOW	NOT FOUND	FIRST PRIORITY
Kenya	Community conservancies beside a county reserve	Two bed caps; US$130 fee; lease history to 2012	Most bed caps; lease renewal terms	Renew leases with escalation tied to the fee pool
Tanzania	State parks and concessions	Entry and per-night concession fees	Bed caps	Publish caps alongside concession fees
Botswana	State concessions	24-bed cap per lodge; fee schedule	Bed count after 2014; concession rents	Publish concession terms and current beds
Namibia	Communal conservancies with JV lodges	Annual income, jobs and area	Beds per conservancy; per-night fees	Add a per-night conservancy fee to JV agreements
Rwanda	Permit-capped park; park partnership	Permits, park revenue, 10% share	Lodge beds around the parks	Keep the permit cap; publish beds per hectare
Uganda	Permit-capped parks	Permit tariff	Revenue by park	Publish park revenue
South Africa	Private reserves beside a national park	Gate and conservation fees	Bed totals	Publish bed totals per reserve
Zambia	State parks	Entry fees (US$25 a day)	Beds; concession terms	Review fees against capacity
Côte d'Ivoire, Senegal, Gabon, Rep. of the Congo	Large state parks; long management agreements	Area; UNESCO status; Odzala agreement	Beds, fees and land payments	Write cap, fee and community share into the first concessions

Source: Heirstone analysis. Assessment from the sources cited in Chapters 2 to 7, including UNESCO WHC (Comoé, Niokolo-Koba, Lopé-Okanda), African Parks (Akagera, Odzala-Kokoua) and Singita (Grumeti). "Not found" means not found in published sources.

CHAPTER 8

08 What each reader should do

TOURISM AUTHORITIES, GOVERNMENTS AND LAND AGENCIES

1 **Collect conservation money on the guest-night.**
Charge it through the camp, as the Mara conservancies (US$130), Ol Pejeta (US$187) and Ngorongoro (US$59) do. Botswana's proposed US$30 arrival levy was withdrawn in the year it was announced, according to Your Botswana.

2 **Publish the cap per hectare.**
Every conservancy, concession and new destination should state beds, permits or keys per 1,000 ha and the area it applies to. We could not find bed totals for Mara North, Sabi Sand or the Serengeti.

3 **Review fees on a fixed timetable.**
Botswana's 2022 review, the first in more than 20 years, raised US$7.8m in year one against US$1.8m projected. Test any rise against capacity: the Mara reserve's 2024 rise coincided with falling visits.

4 **In North Africa, pilot one capped concession per country.**
Choose a site with a draw and controlled access: Wadi El Gemal or Siwa in Egypt, Imlil in Morocco, Ichkeul in Tunisia, Tassili n'Ajjer in Algeria. Write in a bed cap, a term of 15 years or more, a per-night fee, a community levy on the Kasbah du Toubkal model and annual disclosure.

5 **Use concessions as a revenue line.**
Egypt's 3 million reserve visitors are the base: at a tenth of Rwanda's park revenue per visitor they would pay up to about US$84m a year. Concessions, about 3% of protected-area income in 2013, have clear room to grow.

LANDOWNERS AND COMMUNITY BODIES

6 **Tie lease escalation to the fee pool.**
At Olare Motorogi a lease at the 2011 rate equals 14% of the full-occupancy fee pool. A floor rate per hectare plus a share of fees above a threshold lets landowners share in rising prices.

7 **Negotiate the renewal before it is due.**
The Olare Orok contract ran to 2025. Terms of 15 years match the life of a camp; shorter terms raise the operator's risk and lower what it can pay.

8 **Keep the register and the costs in view.**
Management costs equalled a third of lease payments at Olare Orok in 2012, and average holdings range from about 5 to 85 ha per landowner. Registers that reflect households widen the benefit; in one Mara survey a woman was the registered member in 3% of member households.

9 **Document consent before signing carbon contracts.**
The 2025 Isiolo ruling turned on public participation in setting up the conservancies. Keep records of consultation and consent for each participating conservancy before credits are sold.

INVESTORS AND LENDERS

10 Underwrite the lease and the cap.

Value a camp on revenue per hectare and the remaining life of its land right. Match loan tenor to the lease, and check the renewal position before lending.

11 Stress occupancy and currency.

Wilderness earned about US$660 to US$800 per bed-night while occupancy moved between 58% and 67%. Model at least the five-point fall in occupancy it recorded in the year to February 2016, and the exchange rate.

12 Treat carbon as upside.

Carbon paid US$1.30 per hectare in northern Kenya in 2024, and issuance or sales were halted twice between 2023 and 2025. Keep it out of the base case.

LODGE OPERATORS

13 Publish per-night community payments.

The Maa Trust receives US$5 per guest-night from Olare Motorogi camps; Kasbah du Toubkal adds 5% to every invoice. Wilderness's community payments were about 0.8% of turnover in 2012.

14 Count jobs per bed and hire locally.

Wilderness employed about 5.5 people per occupied bed across its group in 2012, including head office and aviation staff. Naboisho reports more than 90 jobs, about 90% filled from surrounding communities.

MASTER DEVELOPERS IN THE GULF AND NORTH AFRICA, AND CONSERVATION BODIES

15 Report hectares per bed alongside the share of land developed.

The Red Sea's caps equal 933 ha per key and AlUla's 2035 plan 240 ha per key. Stating them per bed allows comparison with Mombo (1,750 to 1,944 ha) and the Mara (150 to 162 ha).

16 Add a per-night conservation fee and a land line.

Caps and tenure are already in place in the Saudi destinations; a ring-fenced fee per guest-night would complete the model and give the reserves around the resorts a revenue line tied to use.

17 For conservation bodies: measure per hectare.

Publish income per hectare and beds per 1,000 ha with the annual report, as NACSO and RDB publish their totals, so that governments can compare models.

THE THREE NUMBERS TO ASK FOR, BY MODEL

MODEL	BEDS PER 1,000 HA	FEE PER GUEST	LAND PAYMENT
Community conservancy (Mara)	6.2 to 6.7 beds	US$130 a night (2026)	US$25 to US$45 per ha a year
Communal conservancy (Namibia)	not published	not published	US$0.54 per ha, all benefits (2023)
State concession (Okavango)	1.29 beds (2014); Mombo 0.51 to 0.57	BWP270 (US$22) a day; in Mombo rate	n/a (state land)
Permit-capped park (Volcanoes)	7.4 permits a day	US$1,500 permit	10% of revenue to communities
Developer destination (Saudi)	1.07 to 4.17 keys	not published	not published
Egyptian protected areas	not published	US$3 to US$5 a day (EEAA, 2013)	n/a (state land)

Sources and calculations as in Exhibits 5, 12 and 16.

NOTES, SOURCES AND DEFINITIONS

Definitions and method

Definitions

Bed, guest bed. A sleeping place for one guest. Where operators state maximum guests (Mombo, 18), we use that number and show the berth count where it differs (Mombo, 20). Hotel keys (rooms or suites) are shown separately and usually hold more than one bed.

Beds per 1,000 ha; hectares per bed. Published bed, permit or key cap divided by the stated area of land it applies to. Heirstone analysis.

Conservancy fee. A charge per guest per night, bundled into camp rates in the Mara, that funds leases, management and conservation. Distinct from park entry fees.

Receipts per arrival. International tourism receipts divided by international arrivals (World Bank WDI, from UN Tourism) or national equivalents. Arrivals include business, conference, cross-border and diaspora visitors. Coverage differs by country: some series include same-day visitors, and receipts include international passenger transport where reported.

Revenue or income per hectare. An annual money flow divided by the land area it relates to. The measure is stated in each case: payment to landowners or communities, gross park or operator revenue, or state fee income.

Rack rate. The published price per person per night sharing. Trade and net rates are lower; illustrative calculations from rack rates are ceilings.

Exchange rates

World Bank official exchange rates, annual averages (PA.NUS.FCRF): BWP per US$ 6.84 (2011), 7.64 (2012), 10.90 (2016), 10.20 (2018), 12.37 (2022); N$ 18.45 (2023); ZAR 17.89 (2025); KES 84.53 (2012), 134.82 (2024); EGP 6.06 (2012), 17.78 (2017), 45.30 (2024); MAD 9.94 (2024); TND 3.11 (2024); RWF 1,428 (2025); EUR 0.89 (2019). Wilderness's year to February 2012 uses BWP 6.97, the 2011 and 2012 averages weighted by month. Figures from different years are not adjusted for inflation or exchange-rate changes; fees set in local currency, including Egyptian pounds, are converted at the rate for the year shown and change in dollar terms as rates move.

The Heirstone Bed and Hectare Benchmark

Compiled in 2026 from public sources: conservancy and park-authority websites and tariffs, operator rate pages, UNESCO and IUCN documents, IFC disclosures, NACSO and NRT annual reports, and company results. A site is included where both an area and a bed, permit or key capacity are published, or where the site sets a price or land-payment benchmark. All ratios are Heirstone analysis; inputs are cited with each exhibit.

Conflicts and judgements

Olare Motorogi is listed as 33,386 acres and as 15,200 ha; we use 15,200 ha (also FAO's figure for Olare Orok and Motorogi together), the larger area, which gives the lower density. Naboisho is 22,500 ha (association), 60,000 acres (own site) or 20,000 ha (2018). Volcanoes National Park covers about 13,000 ha before a planned expansion (Afrik21, 2022); we use that area. Namibian per-hectare figures divide conservancy income by the 166,179 km² that conservancies manage; on the 185,809 km² under all community conservation they would be about 11% lower. The N$57.7m joint-venture figure is wages paid to residents, not fees paid to conservancies. The association's current lease total of "approximately 7.5 million" does not state a currency and is not used. Mombo capacity is given as 18 guests by the operator, whose eight double suites and four-berth family unit sleep 20; we show both where it matters. IUCN gives the 2,129 Okavango beds for an area of 16,500 km², and we use that area; on the 2,023,590 ha World Heritage property the density would be 1.05 beds per 1,000 ha. MMWCA's conservancy page gives header figures that differ from its own descriptions for several conservancies; Exhibit 10 uses the header figures.

The "one bed per 700 acres" rule could not be traced to a primary source and is not used as a benchmark. Single-source claims (Tanzania's 2024 earnings, Botswana's levy withdrawal) are attributed in the text. Coordinates on the map are approximate unless sourced from UNESCO or GeoNames.

Fact base

About 170 sourced facts. Exchange rates, francophone park data and the headline figures were checked against World Bank, UNESCO, African Parks, operator and association publications in 2026.

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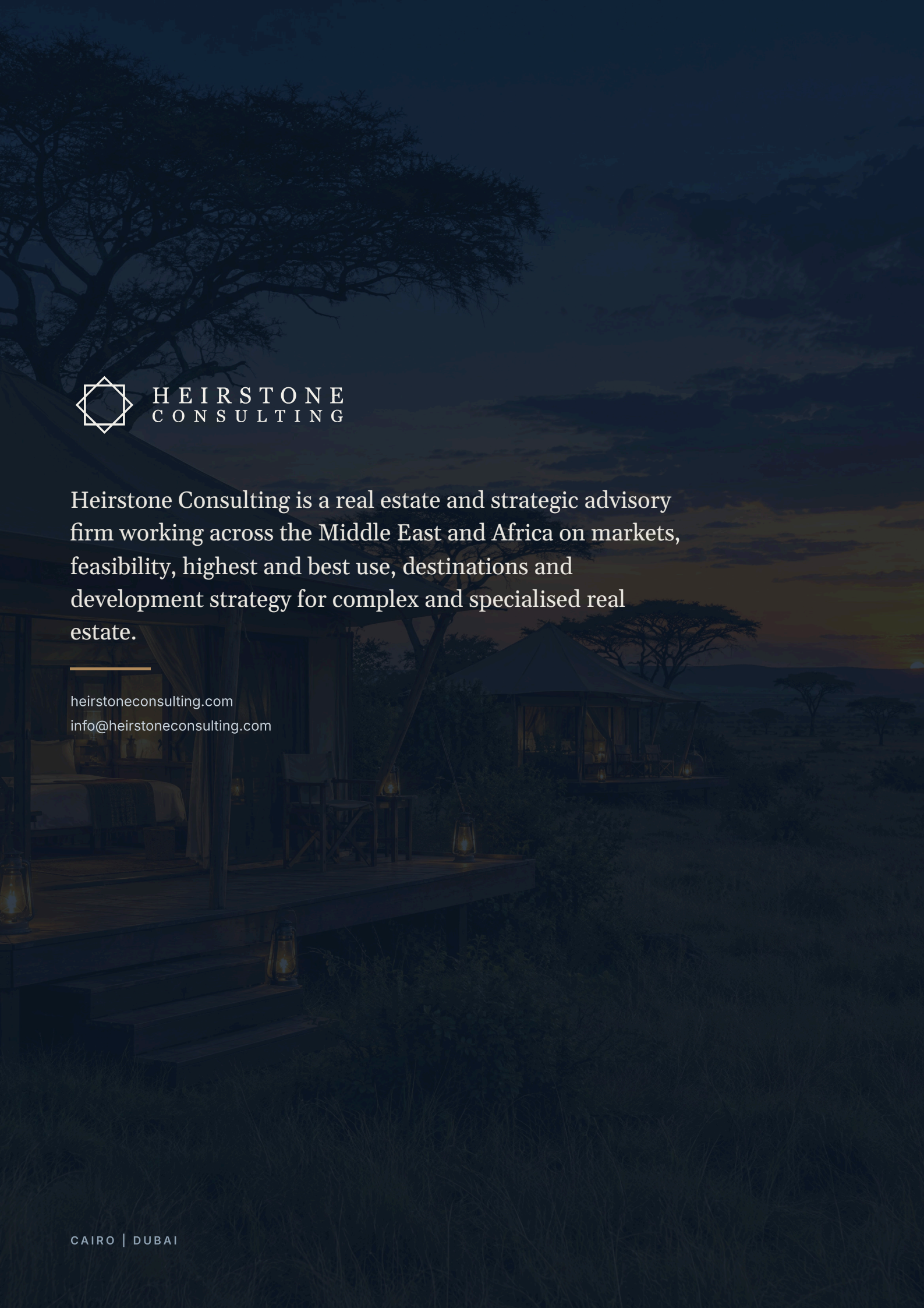
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Chapter 8

Evidence as cited in Chapters 1 to 7. Recommendations are Heirstone's and are not endorsed by any organisation named.

Important notice

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